

ECON 609: Linearization

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Lecture # 2

Basic RBC

- ▶ Benchmark set up:

$$\max \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \{ \log c_t + \psi \log(1 - l_t) \}$$

$$c_t + k_{t+1} = k_t^\alpha (e^{z_t} l_t)^{1-\alpha} + (1 - \delta)k_t, \quad \forall t > 0$$

$$z_t = \rho z_{t-1} + \epsilon_t, \quad \epsilon \sim \mathcal{N}(0, \sigma^2)$$

- ▶ This is a dynamic optimization problem.
- ▶ This problem does not have a "paper and pencil" solution.
- ▶ Traditional solution: linearization.

Equilibrium Conditions

- From the household's problem + firm's problem + aggregate conditions:

$$\frac{1}{c_t} = \beta \mathbb{E}_t \left\{ \frac{1}{c_{t+1}} \left(1 + \alpha k_{t+1}^{\alpha-1} (e^{z_{t+1}} l_{t+1})^{1-\alpha} - \delta \right) \right\}$$

$$\psi \frac{c_t}{1 - l_t} = (1 - \alpha) k_t^\alpha (e^{z_t} l_t)^{1-\alpha} l_t^{-1}$$

$$c_t + k_{t+1} = k_t^\alpha (e^{z_t} l_t)^{1-\alpha} + (1 - \delta) k_t$$

$$z_t = \rho z_{t-1} + \epsilon_t, \quad \epsilon \sim \mathcal{N}(0, \sigma^2)$$

Steady State I

- ▶ If $\sigma = 0$, the equilibrium conditions are:

$$\begin{aligned}\frac{1}{c_t} &= \beta \frac{1}{c_{t+1}} (1 + \alpha k_{t+1}^{\alpha-1} l_{t+1}^{1-\alpha} - \delta) \\ \psi \frac{c_t}{1 - l_t} &= (1 - \alpha) k_t^\alpha l_t^{-\alpha} \\ c_t + k_{t+1} &= k_t^\alpha l_t^{1-\alpha} + (1 - \delta) k_t\end{aligned}$$

- ▶ The equilibrium conditions imply a steady state:

$$\begin{aligned}\frac{1}{c} &= \beta \frac{1}{c} (1 + \alpha k^{\alpha-1} l^{1-\alpha} - \delta) \\ \psi \frac{c}{1 - l} &= (1 - \alpha) k^\alpha l^{-\alpha} \\ c + \delta k &= k^\alpha l^{1-\alpha}\end{aligned}$$

Steady State II

► Solution:

$$k = \frac{\mu}{\Omega + \varphi\mu}$$

$$l = \varphi k$$

$$c = \Omega k$$

$$y = k^\alpha l^{1-\alpha}$$

where $\varphi = \left(\frac{1}{\alpha} \left(\frac{1}{\beta} - 1 + \delta \right) \right)^{\frac{1}{1-\alpha}}$, $\Omega = \varphi^{1-\alpha} - \delta$, and $\mu = \frac{1}{\psi}(1-\alpha)\varphi^{-\alpha}$.

Log-linearization

- ▶ Take variable x_t and substitute by $xe^{\hat{x}_t}$ where:

$$\hat{x}_t = \log \frac{x_t}{x}$$

- ▶ A variable $\hat{x}_t$ represents the log-deviation with respect to the steady state.
- ▶ Linearize with respect to $\hat{x}_t$.

Log-linearization

- ▶ We linearize the equilibrium conditions around l , k , and c with a first-order Taylor expansion.
- ▶ Then we get

$$-\frac{1}{c}(c_t - c) = \mathbb{E}_t \left[\begin{array}{l} -\frac{1}{c}(c_{t+1} - c) + \alpha(1 - \alpha)\beta \frac{y}{k} z_{t+1} + \\ \alpha(\alpha - 1)\beta \frac{y}{k^2}(k_{t+1} - k) + \alpha(1 - \alpha)\beta \frac{y}{kl}(l_{t+1} - l) \end{array} \right]$$

$$\frac{1}{c}(c_t - c) + \frac{1}{1 - l}(l_t - l) = (1 - \alpha)z_t + \frac{\alpha}{k}(k_t - k) - \frac{\alpha}{l}(l_t - l)$$

$$(c_t - c) + (k_{t+1} - k) = \left[\begin{array}{l} y \left((1 - \alpha)z_t + \frac{\alpha}{k}(k_t - k) + \frac{1 - \alpha}{l}(l_t - l) \right) \\ + (1 - \delta)(k_t - k) \end{array} \right]$$

$$z_t = \rho z_{t-1} + \epsilon_t$$

Rewriting the System

- ▶ After some algebra, the system is reduced to

$$A(k_{t+1} - k) + B(k_t - k) + C(l_t - l) + Dz_t = 0$$

$$\mathbb{E}_t \left[\begin{array}{c} G(k_{t+1} - k) + H(k_t - k) + J(l_{t+1} - l) \\ + K(l_t - l) + Lz_{t+1} + Mz_t \end{array} \right] = 0$$

$$\mathbb{E}_t z_{t+1} = \rho z_t$$

- ▶ How do we solve this system of equations?

Undetermined Coefficients

- ▶ We guess policy functions of the form

$$(k_{t+1} - k) = P_1(k_t - k) + P_2 z_t$$

$$(l_t - l) = R_1(k_t - k) + R_2 z_t$$

- ▶ Plug them in, use linearity of expectation, and $\mathbb{E}_t z_{t+1} = \rho z_t$ to get

$$A(P_1(k_t - k) + P_2 z_t) + B(k_t - k) + C(R_1(k_t - k) + R_2 z_t) + D z_t = 0$$

$$G(P_1(k_t - k) + P_2 z_t) + H(k_t - k) + J(R_1(P_1(k_t - k) + P_2 z_t) + R_2 N z_t)$$

$$+ K(R_1(k_t - k) + R_2 z_t) + (LN + M)z_t = 0$$

Solving the System

- ▶ Since these equations need to hold for any value $(k_t - k)$ or z_t , we need to equate each coefficient to zero.
- ▶ Coefficients on $(k_t - k)$:

$$AP_1 + B + CR_1 = 0$$

$$GP_1 + H + JR_1P_1 + KR_1 = 0$$

- ▶ Coefficients on z_t :

$$AP_2 + CR_2 + D = 0$$

$$(G + JR_1)P_2 + JR_2N + KR_2 + LN + M = 0$$

- ▶ Then we can solve for P_1, P_2, R_1 , and R_2 .

Practical Implementation

- ▶ How do we do this in practice?
- ▶ Using dynare in Matlab
- ▶ On solving quadratic equations: "A toolkit for analyzing nonlinear dynamic stochastic models easily" by Harald Uhlig.

General Structure of Linearized System

- ▶ Given m states s_t , n controls y_t , and k exogenous stochastic processes z_{t+1} , we have

$$As_t + Bs_{t-1} + Cy_t + Dz_t = 0$$

$$\mathbb{E}_t(Fs_{t+1} + Gs_t + Hs_{t-1} + Jy_{t+1} + Ky_t + Lz_{t+1} + Mz_t) = 0$$

$$\mathbb{E}_t z_{t+1} = Nz_t$$

where C is of size $l \times n$, $l \geq n$ and of rank n , F is of size $(m + n - l) \times n$, and that N has only stable eigenvalues.

Policy Functions I

- We guess policy functions of the form:

$$s_t = Ps_{t-1} + Qz_t$$

$$y_t = Rs_{t-1} + Uz_t$$

where P , Q , R , and U are matrices such that the computed equilibrium is stable.

Policy Functions II

- For simplicity, suppose $l = n$.

- (1) P satisfies the matrix quadratic equation:

$$(F - JC^{-1}A)P^2 - (JC^{-1}B - G + KC^{-1}A)P - KC^{-1}B + H = 0$$

The equilibrium is stable iff $\max(\text{abs}(\text{eig}(P))) < 1$.

- (2) R is given by $R = -C^{-1}(AP + B)$.

- (3) Q satisfies

$$\begin{aligned} N' \otimes (F - JC^{-1}A) + I_k \otimes (JR + FP + G - KC^{-1}A) \text{vec}(Q) \\ = \text{vec}((JC^{-1}D - L)N + KC^{-1}D - M) \end{aligned}$$

- (4) U satisfies $U = -C^{-1}(AQ + D)$.

Policy Functions III

- ▶ Related to (1), the question becomes how to solve quadratic equations.
- ▶ Procedures to solve quadratic systems: Blanchard and Kahn (1980), Uhlig (1999), Sims (2000), Klein (2000)
- ▶ All of them render equivalent results.

Bayesian Inference

- ▶ Suppose model parameters θ . We want to estimate it.
- ▶ Likelihood function $p(Y|\theta)$ requires that a model is solved numerically given θ .
- ▶ So far, we have seen how to solve a linearized model given θ .
- ▶ Recall: Bayes theorem

$$p(\theta|Y) \propto p(Y|\theta)p(\theta)$$

State-Space Representation and Likelihood

- ▶ Measurement equation:

$$y_t = \Psi_1(\theta) + \Psi_2(\theta)s_t + u_t$$

- ▶ State transition equation:

$$s_t = \Phi_1(\theta)s_{t-1} + \Phi_\epsilon(\theta)\epsilon_t$$

- ▶ Joint density for the observations and latent states:

$$\begin{aligned} p(Y_{1:T}, S_{1:T}|\theta) &= \prod_{t=1}^T p(y_t, s_t | Y_{1:t-1}, S_{1:t-1}, \theta) \\ &= \prod_{t=1}^T p(y_t | s_t, \theta) p(s_t | s_{t-1}, \theta). \end{aligned}$$

- ▶ Problem: we need $p(Y_{1:T}|\theta)$.

Filtering - General Idea

- ▶ Likelihood Function $p(Y_{1:T}|\theta) = \prod_{t=1}^T p(y_t|Y_{1:t-1}, \theta)$.
- ▶ A filter generates a sequence of $s_t|Y_{1:t}$.
- ▶ Iterations:
 - ▶ Initialization at time $t - 1$: $p(s_{t-1}|Y_{1:t-1}, \theta)$
 - ▶ Forecasting t given $t - 1$:

(1) From transition equation:

$$p(s_t|Y_{1:t-1}, \theta) = \int p(s_t|s_{t-1}, Y_{1:t-1}, \theta)p(s_{t-1}|Y_{1:t-1}, \theta)ds_{t-1}$$

(2) From measurement equation:

$$p(y_t|Y_{1:t-1}, \theta) = \int p(y_t|s_t, Y_{1:t-1}, \theta)p(s_t|Y_{1:t-1}, \theta)ds_t$$

- ▶ Updating with Bayes theorem. Once y_t becomes available:

$$p(s_t|Y_{1:t}, \theta) = p(s_t|y_t, Y_{1:t-1}, \theta) = \frac{p(y_t|s_t, Y_{1:t-1}, \theta)p(s_t|Y_{1:t-1}, \theta)}{p(y_t|Y_{1:t-1}, \theta)}$$

Kalman Filter (Linear+Gaussian)

- ▶ Let's consider the state-space model

$$y_t = \Psi s_t + u_t, \quad u_t \sim_{i.i.d} N(0, H)$$

$$s_t = \Phi s_{t-1} + \epsilon_t, \quad \epsilon_t \sim_{i.i.d} N(0, \Sigma)$$

- ▶ Initialization: with stationary s_t , we can initialize the filter with the unconditional distribution of s_t . Covariance matrix:

$$\mathbb{E}[s_t s_t'] = \Phi \mathbb{E}[s_t s_t'] \Phi' + \Sigma$$

- ▶ In linear Gaussian state-space model, all distributions are Gaussian. Thus Kalman filter only tracks means and covariance matrices.

Kalman Filter

(1) Initialization. Prior distribution for initial state s_0 :

$$s_0 \sim N(\bar{s}_{0|0}, P_{0|0}).$$

(2) Recursion. For $t = 1, \dots, T$:

- Forecasting s_t . After observing y_{t-1} , the belief about the state vector has the form $s_{t-1}|Y_{1:t-1} \sim N(\bar{s}_{t-1|t-1}, P_{t-1|t-1})$. Since s_{t-1} and ϵ_t are independent multivariate normal r.v.s,

$$s_t|Y_{1:t-1} \sim N(\bar{s}_{t|t-1}, P_{t|t-1})$$

where

$$\bar{s}_{t|t-1} = \Phi \bar{s}_{t-1|t-1}$$

$$P_{t|t-1} = \Phi P_{t-1|t-1} \Phi' + \Sigma$$

Kalman Filter

► Recursion. (cont)

- Forecasting y_t . The conditional distribution of $y_t|s_t, Y_{1:t-1}$ is of the form $y_t|s_t, Y_{1:t-1} \sim N(\Psi s_t, H)$.

Since $s_t|Y_{1:t-1} \sim N(\bar{s}_{t|t-1}, P_{t|t-1})$, we can deduce that the marginal distribution of y_t conditional on $Y_{1:t-1}$ is of the form

$$y_t|Y_{1:t-1} \sim N(\bar{y}_{t|t-1}, F_{t|t-1})$$

where $\bar{y}_{t|t-1} = \Psi \bar{s}_{t|t-1}$, $F_{t|t-1} = \Psi P_{t|t-1} \Psi' + H$.

- Likelihood function:

$$P(Y_{1:T}|\Psi, \Phi, \Sigma, H) = (2\pi)^{-nT/2} \left(\prod_{t=1}^T |F_{t|t-1}| \right)^{-1/2} \\ \times \exp \left\{ -\frac{1}{2} \sum_{t=1}^T (y_t - \bar{y}_{t|t-1})' F_{t|t-1}^{-1} (y_t - \bar{y}_{t|t-1}) \right\}$$

Kalman Filter

- Recursion. (cont)

- Updating. Applying Bayes theorem, we obtain

$s_t | y_t, Y_{1:t-1} \sim N(\bar{s}_{t|t}, P_{t|t})$ where

$$\bar{s}_{t|t} = \bar{s}_{t|t-1} + P_{t|t-1} \Psi' F_{t|t-1}^{-1} (y_t - \bar{y}_{t|t-1})$$

$$P_{t|t} = P_{t|t-1} - P_{t|t-1} \Psi' F_{t|t-1}^{-1} \Psi P_{t|t-1}$$

- This complete one iteration of the algorithm. The posterior $s_t | Y_{1:t}$ is the prior for the next iteration.

Summary: Conditional Distributions for Kalman Filter

	Distribution	Mean and Variance
$s_{t-1} (Y_{1:t-1}, \theta)$	$N(\bar{s}_{t-1 t-1}, P_{t-1 t-1})$	Given from Iteration $t - 1$
$s_t (Y_{1:t-1}, \theta)$	$N(\bar{s}_{t t-1}, P_{t t-1})$	$\bar{s}_{t t-1} = \Phi \bar{s}_{t-1 t-1}$ $P_{t t-1} = \Phi P_{t-1 t-1} \Phi' + \Sigma$
$y_t (Y_{1:t-1}, \theta)$	$N(\bar{y}_{t t-1}, F_{t t-1})$	$\bar{y}_{t t-1} = \Psi \bar{s}_{t t-1}$ $F_{t t-1} = \Psi P_{t t-1} \Psi' + H$
$s_t (Y_{1:t}, \theta)$	$N(\bar{s}_{t t}, P_{t t})$	$\bar{s}_{t t} = \bar{s}_{t t-1} + P_{t t-1} \Psi' F_{t t-1}^{-1} (y_t - \bar{y}_{t t-1})$ $P_{t t} = P_{t t-1} - P_{t t-1} \Psi' F_{t t-1}^{-1} \Psi P_{t t-1}$

What To Do

- ▶ We have a prior $p(\theta)$.
- ▶ We are able to
 - ▶ solve the model
 - ▶ cast it into a state-space form
 - ▶ evaluate the likelihood function
- ▶ Now we need a posterior simulator to generate draws for θ .

Random-Walk Metropolis Algorithm for a Given Model

- (1) Our objective function is the posterior, $p(Y|\theta)p(\theta)$.
- (2) Draw $\theta^{(0)}$ from $N(\tilde{\theta}, c_0^2 \tilde{\Sigma})$ or directly specify a starting value.
- (4) For $s = 1, \dots, n_{sim}$: draw θ^* from the proposal $N(\theta^{(s-1)}, c^2 \tilde{\Sigma})$.
The jump from $\theta^{(s-1)}$ is accepted ($\theta^{(s)} = \theta^*$) with probability $\min\{1, r(\theta^{(s-1)}, \theta^* | Y)\}$ and rejected ($\theta^{(s)} = \theta^{(s-1)}$) otherwise.

$$r(\theta^{(s-1)}, \theta^* | Y) = \frac{p(Y|\theta^*)p(\theta^*)}{p(Y|\theta^{(s-1)})p(\theta^{(s-1)})}$$